ATM Error Codes and Troubleshooting Guide

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Department: ATM Operations & Technical Support

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Cash Dispenser Errors {#cash-dispenser-errors}

Error Code: CD-001 - Cash Not Dispensing

Priority: CRITICAL

Symptoms: ATM accepts transaction but no cash is dispensed

Root Causes:

- Empty cash cassette
- Jammed bills in dispenser mechanism
- Faulty solenoid valve
- Sensor malfunction

Troubleshooting Steps:

1. Check cash cassette levels immediately
2. Open dispenser module and inspect for jammed bills
3. Remove any stuck currency carefully using anti-static tools
4. Test solenoid valve operation (should hear clicking sound)
5. Clean optical sensors with isopropyl alcohol and lint-free cloth
6. Run diagnostic test mode: Press F1+F3 simultaneously at boot
7. Check error log for sensor failure codes (CD-001A, CD-001B)
8. If issue persists, replace dispenser motor assembly

Parts Required:

- Replacement bills picker assembly (Part #: DP-2847)
- Anti-static cleaning kit
- Optical sensor cleaning solution

Average Resolution Time: 15-30 minutes
Downtime Impact: HIGH - ATM must be taken offline

Error Code: CD-002 - Double Note Detection

Priority: HIGH
Symptoms: Multiple bills dispensed for single withdrawal

Root Causes:

- Worn separation rollers
- Incorrect tension settings
- Static electricity buildup
- Damaged currency notes

Troubleshooting Steps:

1. Access dispenser settings menu (Admin > Diagnostics > Dispenser)
2. Reduce roller tension by 5% increments
3. Clean separation rollers with denatured alcohol
4. Check for bent or damaged currency in cassette
5. Replace separation roller if worn (check for flat spots)
6. Verify double-detect sensor calibration
7. Test with fresh currency stack

Preventive Maintenance:

- Replace separation rollers every 6 months

- Clean double-detect sensors weekly
- Maintain humidity levels between 40-60%

Average Resolution Time: 20-40 minutes

Error Code: CD-003 - Cash Jam Detected

Priority: CRITICAL
Symptoms: Bills stuck in transport path, "Out of Service" displayed

Immediate Actions:

1. Press emergency stop button (red button on dispenser)
2. Place ATM in out-of-service mode
3. Document time and transaction number
4. Open dispenser access panel with master key

Clearing Procedure:

1. Locate jam position using diagnostic display
2. Carefully remove jammed bills following transport path direction
3. Check for torn currency fragments
4. Inspect transport belts for wear or debris
5. Clean all transport rollers with approved cleaning solution
6. Reset dispenser error counter: Menu > Service > Reset Errors
7. Run empty cycle test (no cash) 5 times

8. Load test cassette and perform test dispense

Post-Resolution:

- Document removed cash amount for reconciliation
- Update maintenance log
- Monitor next 10 transactions closely

Average Resolution Time: 30-45 minutes

Card Reader Errors {#card-reader-errors}

Error Code: CR-101 - Card Read Error

Priority: HIGH
Symptoms: "Cannot read card" message, card rejected immediately

Root Causes:

- Dirty magnetic stripe reader
- Worn read head
- Damaged customer card
- Incorrect card insertion

Troubleshooting Steps:

1. Clean card reader head using approved cleaning card (Part #: CC-9901)
2. Run cleaning card through reader 3 times
3. Test with known good card (test card in ATM toolkit)

4. Check reader head alignment using gap gauge tool
5. Inspect card entry path for obstructions
6. Verify reader voltage: Should be 12V DC ±0.5V
7. Check connector pins for corrosion
8. If persistent, replace card reader module

Cleaning Schedule:

- Daily: Visual inspection
- Weekly: Cleaning card procedure
- Monthly: Deep clean with isopropyl alcohol

Average Resolution Time: 10-15 minutes

Error Code: CR-102 - Card Retained/Captured

Priority: CRITICAL
Symptoms: Customer card stuck inside reader, cannot retrieve

Emergency Procedure:

1. Inform customer card will be retrieved and secured
2. Note customer details and transaction ID
3. Access card capture bin (requires supervisor key)
4. Document card details: Bank, last 4 digits, customer info
5. Store card in secure envelope with incident report
6. Complete card retention form CR-102F

Technical Resolution:

1. Check card retention counter: Admin > Statistics > Captures
2. If count > 50, empty capture bin immediately
3. Inspect retention mechanism for obstructions
4. Test card return motor operation
5. Calibrate card positioning sensor
6. Update card reader firmware if version < 4.8.2

Customer Service:

- Provide card retention receipt
- Advise customer to contact bank within 24 hours
- Card held for 30 days per regulatory requirements

Average Resolution Time: 15-20 minutes

Error Code: CR-103 - Card Reader Motor Failure

Priority: CRITICAL
Symptoms: Card not moving in/out, grinding noise

Diagnosis:

1. Listen for motor operation (should be smooth)
2. Check motor power supply: 12V DC
3. Test motor independently using diagnostic tool

4. Inspect drive belt for wear or breaks
5. Check motor mounting screws for looseness

Replacement Procedure:

1. Power down ATM completely
2. Disconnect card reader harness (8-pin connector)
3. Remove 4 mounting screws
4. Replace card reader module (Part #: CRM-5544)
5. Reconnect harness ensuring proper pin alignment
6. Power up and run initialization sequence
7. Perform 10 test card reads

Average Resolution Time: 25-35 minutes
Parts Cost: $280-350

Receipt Printer Errors {#receipt-printer-errors}

Error Code: RP-201 - Paper Out

Priority: MEDIUM
Symptoms: "Cannot print receipt" message

Resolution Steps:

1. Open printer access door with printer key
2. Remove empty paper roll spindle

3. Load new thermal paper roll (80mm x 80mm)
4. Ensure paper feeds from bottom of roll
5. Thread paper through paper guides
6. Close printer door until click sound
7. Press printer test button (green button)
8. Verify test receipt prints clearly

Paper Specifications:

- Type: Thermal paper, BPA-free
- Width: 80mm (±0.5mm)
- Diameter: Maximum 80mm
- Core: 12mm internal diameter
- Paper grade: Premium thermal 55-65 GSM

Average Resolution Time: 5 minutes

Error Code: RP-202 - Paper Jam

Priority: HIGH
Symptoms: Paper stuck, printer error light flashing red

Clearing Steps:

1. Open printer top cover
2. Locate jammed paper (usually at cutter assembly)
3. Rotate green paper release knob clockwise

4. Gently pull paper in direction of paper path
5. Remove any torn paper fragments
6. Clean cutter blade with soft brush
7. Check paper path for adhesive residue
8. Clean rubber feed rollers
9. Close cover and test print

Prevention:

- Use only approved paper stock
- Store paper in climate-controlled environment
- Replace paper before completely empty
- Clean printer monthly

Average Resolution Time: 10-15 minutes

Error Code: RP-203 - Print Head Overheating

Priority: MEDIUM
Symptoms: Faded prints, "Printer cooling" message

Causes:

- Continuous high-volume printing
- Poor ventilation around printer
- Ambient temperature too high
- Worn thermal print head

Resolution:

1. Allow 10-minute cooling period
2. Check ATM ambient temperature (should be < 30°C)
3. Verify cooling fan operation
4. Clean print head with cleaning pen (Part #: CPH-401)
5. Reduce print density: Settings > Printer > Density (decrease by 10%)
6. Check for printer firmware updates
7. If persistent, schedule print head replacement

Thermal Management:

- Monitor transaction volume during peak hours
- Ensure minimum 6 inches clearance around ATM
- Service HVAC system quarterly

Average Resolution Time: 15-20 minutes

Network and Communication Errors {#network-errors}

Error Code: NET-301 - Host Connection Lost

Priority: CRITICAL
Symptoms: "Cannot connect to bank" message, all transactions decline

Immediate Actions:

1. Check network indicator lights on ATM controller
2. Verify physical cable connections (RJ-45 connectors)
3. Test network connectivity: Diagnostics > Network Test
4. Ping gateway: Should respond < 50ms
5. Check router/switch status lights

Troubleshooting:

1. Restart ATM communication module: Service > Restart Comms
2. Check IP configuration: Should match site documentation
3. Verify DNS settings (Primary: 8.8.8.8, Secondary: 8.8.4.4)
4. Test with backup communication path (if available)
5. Check firewall settings (Port 443 HTTPS must be open)
6. Contact network operations center (NOC) if issue persists

Network Requirements:

- Minimum bandwidth: 256 Kbps
- Maximum latency: 200ms
- Uptime SLA: 99.9%

Escalation: If not resolved in 15 minutes, contact NOC at 1-800-ATM-HELP

Average Resolution Time: 10-30 minutes

Error Code: NET-302 - Encryption Key Error

Priority: CRITICAL
Symptoms: "Security error" message, cannot process transactions

Security Protocol:

1. DO NOT attempt unauthorized key injection
2. Contact security operations immediately
3. Document error time and any suspicious activity
4. Place ATM in supervisor mode (requires dual authentication)

Authorized Resolution:

1. Verify current key status: Admin > Security > Key Status
2. Check key expiration date
3. Initiate remote key injection from secure key management system
4. Verify key injection successful (green indicator)
5. Test transaction processing
6. Document key update in security log

Compliance:

- Only certified technicians may handle key management
- PCI-DSS compliance required
- Dual control policy mandatory

Average Resolution Time: 20-30 minutes
Requires: Level 3 security clearance

Error Code: NET-303 - Timeout Error

Priority: HIGH
Symptoms: Transactions taking unusually long, eventual timeout

Diagnosis:

1. Check network latency: Ping host server
2. Monitor bandwidth usage
3. Check for packet loss: Should be < 1%
4. Verify QoS settings on router
5. Test during off-peak hours

Resolution:

1. Increase timeout values: Config > Timeouts > Host Response (60s → 90s)
2. Clear communication cache: Service > Clear Cache
3. Restart communication service
4. Update network firmware if outdated
5. Contact ISP if issue persists

Performance Benchmarks:

- Authorization response: < 3 seconds
- Balance inquiry: < 2 seconds
- Network round-trip: < 200ms

Average Resolution Time: 15-25 minutes

Display and Touch Screen Errors {#display-errors}

Error Code: DS-401 - Touch Screen Not Responding

Priority: HIGH
Symptoms: Screen visible but touch input not working

Quick Fixes:

1. Clean screen with microfiber cloth
2. Check for moisture or liquid on screen
3. Restart ATM touch screen service
4. Calibrate touch screen: Settings > Display > Calibration

Detailed Troubleshooting:

1. Access touch screen diagnostics: Hold F4 during boot
2. Run touch test pattern (grid should respond to all touches)
3. Check USB connection to touch controller
4. Verify touch controller power: 5V DC
5. Test with stylus to isolate hardware vs. software issue
6. Update touch screen drivers if available
7. Replace touch overlay if physically damaged

Calibration Procedure:

1. Press calibration targets in sequence

2. Tap center of each crosshair precisely
3. Complete 9-point calibration
4. Test accuracy across entire screen

Average Resolution Time: 15-20 minutes

Error Code: DS-402 - Display Backlight Failure

Priority: HIGH
Symptoms: Screen very dim or completely dark, but ATM operational

Causes:

- Failed backlight inverter
- Burned out LED backlight strips
- Loose cable connection
- Power supply issue

Resolution Steps:

1. Check brightness settings: Max brightness to verify
2. Inspect backlight cable connection (4-pin connector)
3. Measure backlight voltage: Should be 12V DC
4. Listen for high-pitched inverter whine (indicates failure)
5. Replace backlight inverter board (Part #: BLI-7723)
6. If inverter OK, replace LCD panel assembly

Safety Warning:

- High voltage present in backlight inverter
- Disconnect power before servicing
- Use insulated tools only

Average Resolution Time: 30-45 minutes
Parts Cost: $150-400

Error Code: DS-403 - Screen Artifacts/Distortion

Priority: MEDIUM
Symptoms: Lines, flickering, or color distortion on display

Troubleshooting:

1. Check video cable connections (VGA or HDMI)
2. Reseat video cable at both ends
3. Test with known good cable
4. Check for electromagnetic interference sources
5. Verify video output settings: 1024x768 @ 60Hz
6. Update graphics drivers
7. Replace LCD panel if hardware defect confirmed

Average Resolution Time: 20-30 minutes

Security and Alarm Errors {#security-errors}

Error Code: SEC-501 - Door Ajar Alert

Priority: CRITICAL
Symptoms: Audible alarm, "Security breach" message

Immediate Response:

1. Verify all doors and panels are closed
2. Check door sensor alignment
3. Inspect for forced entry attempts
4. Review security camera footage
5. Document incident time and location

Technical Resolution:

1. Test each door sensor: Should show "CLOSED" status
2. Adjust sensor mounting if misaligned
3. Clean sensor contacts
4. Replace faulty door sensor (Part #: DS-8801)
5. Reset alarm system: Security > Clear Alarms (requires supervisor code)

Security Protocol:

- Contact security immediately for any actual breach
- Do not reset alarm until area secured
- File incident report within 1 hour

Average Resolution Time: 10-20 minutes

Error Code: SEC-502 - Vibration/Tilt Sensor Alert

Priority: CRITICAL
Symptoms: Alarm triggered, possible physical attack

Emergency Actions:

1. DO NOT approach ATM if attack in progress
2. Contact law enforcement immediately
3. Contact security operations center
4. Monitor via security cameras from safe location
5. Preserve all evidence

Post-Incident:

1. Inspect ATM for physical damage
2. Check cash cassette security
3. Review all sensor logs
4. Test vibration sensor sensitivity
5. Adjust sensitivity if false alarms: Security > Sensors > Vibration (reduce by 10%)

Average Resolution Time: 15-30 minutes
Requires: Security clearance and police clearance

Error Code: SEC-503 - Skimming Device Detected

Priority: CRITICAL
Symptoms: Card reader tamper alert, anti-skimming sensor triggered

Critical Response:

1. Immediately place ATM out of service
2. Contact fraud prevention team
3. Contact law enforcement
4. Do not remove suspected device (evidence preservation)
5. Secure area and prevent customer access

Investigation:

1. Document device location and appearance
2. Photograph from multiple angles
3. Check for camera devices
4. Review transaction logs for affected cards
5. Coordinate with bank security for customer notification

Prevention:

- Weekly physical inspection of card reader
- Anti-skimming detection systems mandatory
- Tamper-evident seals on all access points

Average Resolution Time: 1-2 hours
Requires: Law enforcement coordination

Cash Cassette Errors {#cash-cassette-errors}

Error Code: CC-601 - Cassette Not Detected

Priority: HIGH
Symptoms: "Cash cassette error" message

Resolution:

1. Open safe door with dual key system
2. Remove and reseat cash cassette
3. Ensure cassette fully inserted (should hear click)
4. Check cassette connector pins for damage
5. Verify cassette type matches ATM model
6. Clean cassette connector with contact cleaner
7. Test cassette detection: Diagnostics > Cassette Test

Cassette Specifications:

- Type: 4-denomination cassette
- Capacity: 2,000 notes per denomination
- Connector: 16-pin standard
- Compatible models: Series 2000, 3000, 4000

Average Resolution Time: 15 minutes

Error Code: CC-602 - Cash Low Warning

Priority: MEDIUM
Symptoms: "Low cash" alert, still operational

Actions:

1. Check current cash levels: Admin > Status > Cash Levels
2. Schedule immediate replenishment if < 20% capacity
3. Notify cash services for refill
4. Monitor transaction patterns
5. Update cash forecasting system

Cash Management:

- Target level: 70-90% capacity
- Minimum operating level: 15%
- Emergency refill threshold: 10%
- Average daily dispense: $15,000-25,000

Average Resolution Time: N/A (Requires cash replenishment)

Error Code: CC-603 - Cassette Full (Deposits)

Priority: HIGH

Symptoms: Cannot accept deposits, "Deposit function unavailable"

Resolution:

1. Check deposit cassette capacity
2. Schedule immediate collection if > 90% full
3. Contact CIT (Cash In Transit) services
4. Monitor deposit acceptance rate
5. Temporarily disable deposit function if necessary

Average Resolution Time: N/A (Requires CIT collection)

Power and Hardware Errors {#power-errors}

Error Code: PWR-701 - UPS Battery Low

Priority: HIGH
Symptoms: "Battery backup low" warning

Actions:

1. Verify UPS status lights (should show green)
2. Check UPS battery charge level: Should be > 50%
3. Ensure AC power connected properly
4. Test UPS by unplugging AC briefly (ATM should stay on)
5. If battery not charging, replace UPS battery

UPS Specifications:

- Type: Online double-conversion UPS
- Capacity: 1500VA / 1000W
- Battery: 12V 9Ah sealed lead-acid
- Runtime: 15-20 minutes at full load
- Battery life: 3-5 years

Replacement Procedure:

1. Ensure AC power connected
2. Open UPS access panel
3. Disconnect old battery (note polarity)
4. Install new battery (Part #: UPS-BAT-12V9)
5. Close panel and verify charging

Average Resolution Time: 20-30 minutes
Parts Cost: $60-80

Error Code: PWR-702 - Main Power Failure

Priority: CRITICAL
Symptoms: ATM on UPS power, countdown timer displayed

Immediate Actions:

1. Check main circuit breaker
2. Verify building power status
3. Check ATM power cable and connections

4. Test power outlet with voltmeter (should be 110-120V AC)
5. Contact facilities management if building power issue

Graceful Shutdown:

1. If power not restored within 10 minutes, initiate shutdown
2. Complete any in-progress transactions
3. Place "Out of Service" sign
4. Power down following proper sequence
5. Document downtime in maintenance log

Average Resolution Time: Varies (depends on power restoration)

Error Code: PWR-703 - Cooling Fan Failure

Priority: MEDIUM
Symptoms: Overheating warning, high temperature alert

Resolution:

1. Check fan operation (should hear fan noise)
2. Clean fan blades and vents with compressed air
3. Verify fan power connection
4. Measure temperature: Should be < 35°C internal
5. Replace cooling fan if failed (Part #: FAN-120MM)

Temperature Management:

- Operating range: 10°C to 35°C
- Optimal: 18°C to 25°C
- Critical shutdown: > 45°C

Average Resolution Time: 15-25 minutes

Preventive Maintenance Schedule

Daily Checks

- Visual inspection of exterior
- Check for error messages
- Verify cash levels
- Test basic transaction

Weekly Maintenance

- Clean card reader with cleaning card
- Clean touch screen
- Check receipt paper level
- Inspect for tampering

Monthly Maintenance

- Deep clean all components
- Test UPS battery

- Inspect cable connections
- Update transaction logs
- Clean cooling vents

Quarterly Maintenance

- Replace air filters
- Inspect security seals
- Update firmware
- Full diagnostic test
- Preventive parts replacement

Annual Maintenance

- Replace separation rollers
- Replace receipt printer cutter
- UPS battery replacement (if needed)
- Complete security audit
- Calibration of all sensors

Emergency Contact Numbers

Technical Support: 1-800-ATM-TECH (1-800-286-8324)
Security Operations: 1-800-ATM-SEC1 (1-800-286-7321)
Network Operations: 1-800-NET-OPS1 (1-800-638-6771)
Cash Services: 1-800-CASH-CIT (1-800-227-4248)

Emergency After Hours: +1-555-0199 (24/7)

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- v4.1 (Jul 2025): Updated error codes for new models
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